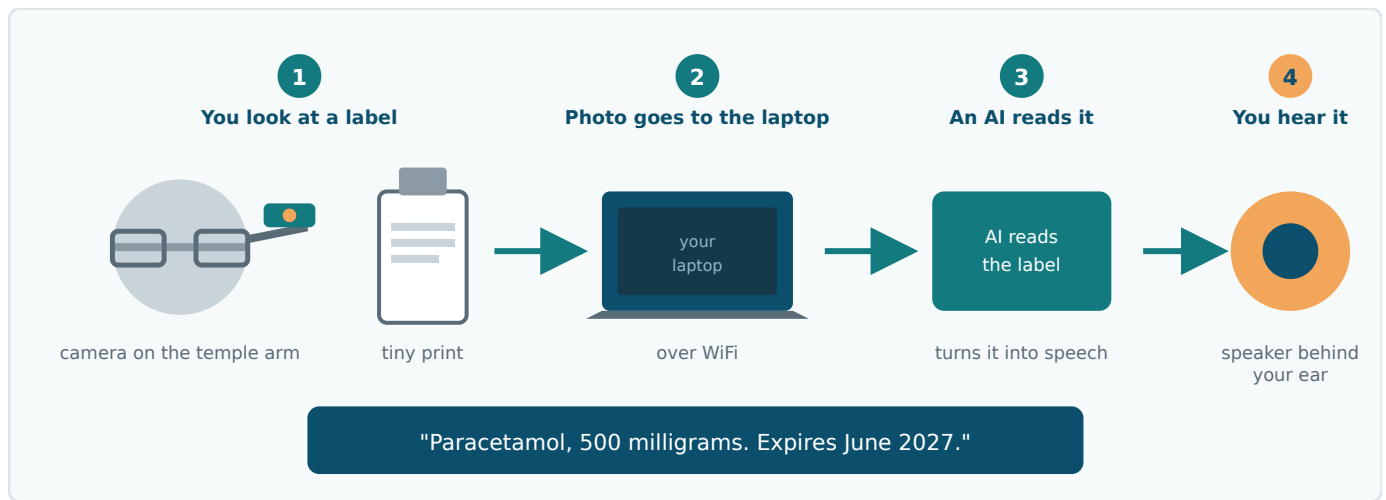


What Am I Building?

The whole thing, from zero, no assumed knowledge

IN ONE SENTENCE

You clip two small plastic boxes onto the arm of a pair of glasses you already own. One box has a camera. The other has a speaker. You tap it, it photographs whatever you're looking at, and it reads the writing out loud to you.



The entire product. Four steps, about five seconds.

The board on your desk is not clever. It only takes a photo, sends it, and plays the sound that comes back. All the thinking happens on your laptop. That's deliberate — it means you can fix and improve the smart part without ever re-flashing the hardware.

WHY GLASSES AND NOT A HANDHELD GADGET

Because people put handheld things down and lose them. Glasses are already on your face — you can't misplace the thing you're wearing. That's the entire product thesis in one line, and it's what you say on stage.

THE THREE THINGS THAT MAKE IT MORE THAN A CAMERA

Feature	What it does for the person wearing it
It reads barcodes first	If there's a barcode, it looks the product up in a database instead of asking the AI. Faster, free, and more accurate on packaged goods.
It remembers a little about you	Either your dosage-relevant facts, or things you asked it to remember. Which one depends on a single setting.
It admits when it can't see	On a bad photo it says "I can't see that clearly" rather than guessing. For someone who can't check the answer, that matters more than being usually right.

1

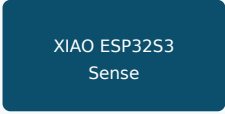
The parts, and what each one does

Every physical thing in the build

FRONT POD — near the hinge



CAMERA
the eye



THE BOARD
XIAO ESP32S3
Sense
camera + WiFi



TOUCH PAD
a coin you tap

This pod SEES. It takes the picture and sends it over WiFi.

48 × 10 × 29 mm

REAR POD — behind the ear



THE AMP
AMPLIFIER
makes it loud enough



SPEAKER
the mouth

This pod SPEAKS. It plays the answer the laptop sends back.

57 × 10 × 38 mm

Two boxes. One sees, one speaks. Five wires join them along the arm.

WHY TWO BOXES INSTEAD OF ONE

If you put all five components in one box it becomes a big lump sitting on the side of your head. Splitting them means each box is small and thin. The camera has to be at the **front** (that's where you look) and the speaker wants to be near your **ear**. So the split matches where things naturally belong.

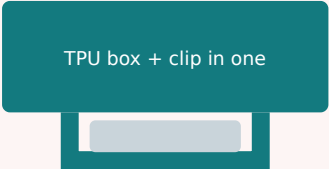
Wire colour	What it carries	In plain words
Red	3.3 volts of power	Feeds the amp
Black	Ground	The return path — every circuit needs one
Yellow	Bit clock	Three wires that together carry digital sound from the board to the amp
Green	Word select	
Blue	Data	
White	Touch signal	Stays inside the front pod — tells the board you tapped it

Don't worry about what "bit clock" means. Those three wires are a standard called I2S. You never have to understand it — you just have to connect yellow-to-yellow, green-to-green, blue-to-blue. The code handles the rest.

WHAT WE HAD BEFORE

The box and the clip were **the same part**. The TPU box had a C-shaped channel moulded into its underside, sized to grip one exact thickness of temple arm. Change glasses, and the box no longer fits — you reprint the whole box.

OLD — box IS the clip

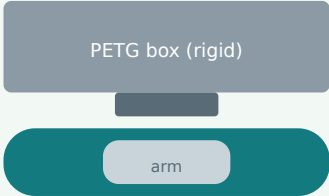


channel sized to ONE arm thickness

wrong size → reprint the whole box

3-4 hours per attempt

NEW — your sketch



ring squishes onto ANY arm

wrong size → reprint a 10-minute ring

The change: stop asking one part to do two jobs.

This is the real win, and it's worth saying out loud in the pitch. The box never touches the glasses. The ring does all the gripping. So the expensive part — the one with electronics glued into it — never has to change when the glasses change. You print rings, not pods.

WHY THE TWO DIFFERENT PLASTICS

Material	Feels like	Used for, and why
TPU	A rubber shoe sole — bends, springs back	The ring. It has to stretch over the arm and squeeze. Rigid plastic would snap or stay loose; TPU grips a range of sizes and survives being pulled on and off hundreds of times.
PETG	A fizzy-drink bottle — firm, slightly tough	The box. It must hold its shape so components don't rattle and the lid stays shut. It never flexes, so it never fatigues.
PLA	Hard and a bit brittle	Test prints only. Fastest and cheapest to print — use it while you're iterating, not in the final build.

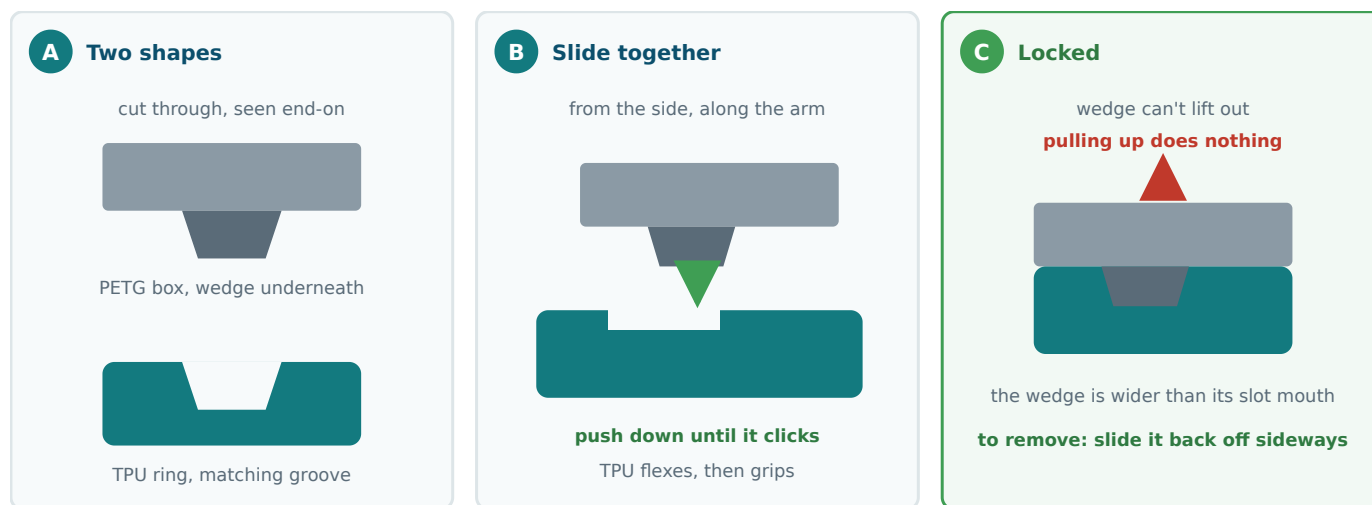
Soft and rigid is the whole trick. Every clip-on product works this way — the springy part grips, the stiff part holds the electronics. If you only take one idea from this page, take that one.

WHAT THE RING BUYS YOU, IN PRACTICE

Situation	Old design	Ring design
A judge wants to try it on their own glasses	Won't fit	Slide on a spare ring
The clip is slightly too loose	Reprint the pod (3-4 hrs)	Reprint a ring (10 min)
A teammate wants to test-wear it	Only if same glasses	Their own ring, same pod
The mount cracks the night before	Electronics stuck in a broken pod	Swap the ring, pod untouched

3 The beam interlock, step by step The last row is the one that should sell you on it. With the old design the thing that flexes most is the thing your electronics are glued into. The ring design moves all the flexing into a part that costs ten minutes to replace.

An interlock is just **a shape that slides into a matching shape and then can't be pulled straight out**. Think of a Lego brick, or a drawer runner. Ours is a wedge — wide at the top, narrow at the bottom.



A wedge in a matching groove. Slides on sideways, won't pull off upwards.

Why a wedge and not a straight peg? A straight peg only holds by friction, and friction on a printed part fades as it wears. A wedge holds *mechanically* — the shape itself blocks the exit. It stays tight even when it's worn smooth.

WHY THIS DIRECTION MATTERS

The slide-on direction runs **along the temple arm**, front to back. That's deliberate: gravity and knocks push the pod **down and outward**, and the interlock resists exactly those. The only direction that releases it is the one nothing pushes it in by accident.

The one number to get right: the gap between wedge and groove. Too tight and it won't go on; too loose and the pod wobbles. Start at **0.3 mm** of clearance — and because the ring is TPU, it forgives a lot. That forgiveness is exactly why the soft part is the one with the groove.

ABOUT THE WORD "FOAM"

Your sketch says *foam ring*. There are two ways to get that, and they're both fine — worth knowing the difference before you print:

Approach	What it means
Printed soft TPU (recommended)	Print the ring in TPU at low infill with a soft pattern. It comes off the printer already squishy. One part, nothing to glue, nothing to fall off.
TPU ring + stick-on foam	Print a thinner TPU ring and line the inside with a strip of adhesive foam tape. Grips even harder and is more forgiving of odd-shaped arms — but the tape is one more thing that can peel off mid-demo.

Start with printed TPU. Add foam tape only if the ring turns out to slip on the actual glasses. Fewer parts is fewer things that fail on stage.

Important, and I want to be straight with you: the STL files sitting in hardware/cad/ right now are the **OLD** design — box-with-built-in-clip. They do not have a ring or an interlock. Nothing on pages 3 and 4 of this document exists as a printable file yet.

Thing	Status	What that means for you
Server + AI	Done	Works today, no hardware needed. Test it now.
Firmware skeleton	Done	Wake word still stubbed — that's planned, not broken.
Old pod STLs	Printable	Will work, but only on one arm size, and needs fit-test tuning.
Ring + interlock	Not built	Your sketch. Needs new CAD before it can be printed.

SO WHAT DO YOU DO ABOUT IT

You have two sane options. Neither is wrong — it depends on how much time you have before demo night.

OPTION A — print the old design now

Gets a physical thing on your face today.

You already have the files.

Cost: fit-test tuning, one arm size only

Best if demo night is close

and you need something that definitely exists

OPTION B — build the ring design

I write the new CAD, you print rings.

Fits any glasses, including a judge's.

Cost: new CAD, new test prints

Best if you have a few days

and it's the better story on stage

You can also do both — print the old pods to keep testing while the new CAD is written.

My honest recommendation: do both, in that order. Print the old design this week so you always have a working physical prototype to fall back on. Then build the ring version as the upgrade. A hackathon is won by the team that has *something that works*, not the team with the best design that didn't get printed in time.

Nothing you've done is wasted either way. The electronics, the wiring, the server, the box dimensions — all identical between the two. The only thing that changes is how the box attaches to the glasses.



Every word explained

Nothing here is assumed knowledge

Word	What it actually means
Pod	One of the two small printed boxes. Front pod = camera. Rear pod = speaker.
Temple arm	The bit of the glasses that goes over your ear.
XIAO ESP32S3 Sense	The little computer board. Has WiFi and a camera socket. About the size of a stamp.
Firmware	The program that runs <i>on the board</i> (as opposed to on your laptop).
Flash	To copy the firmware onto the board over USB. "Flashing the board" = installing the program.
Server	The program running on your laptop that receives photos and answers them.
Endpoint	One specific thing the server can do, named like a web address. <code>/read_label</code> is an endpoint.
curl	A command that sends something to the server from your terminal. Lets you test with no hardware.
I2S	The three-wire standard for sending digital sound. Yellow, green, blue.
Amp / amplifier	Turns a weak digital sound signal into one strong enough to drive a speaker.
Capacitive touch	The board can sense a finger near a metal pad. No button needed — just a wire and a coin.
PSRAM	Extra memory the camera needs. Must be set to "OPI PSRAM" or the camera won't start.
Wake word	Saying "Hey Vocalens" to trigger it instead of tapping. Hardest part — saved for last.
STL	A 3D model file. What you hand to the printer software.
OpenSCAD / <code>.scad</code>	Where the 3D model is <i>written as code</i> . Change a number, re-export the STL.
Slicer	Software that turns an STL into printer instructions. You set layer height and material here.
TPU / PETG / PLA	Rubbery / firm / hard-and-brittle. See page 3.
Interlock	A shape that slides into a matching shape and then can't be pulled out.
Fit test	A tiny 3-minute print used only to check the clip grips before printing a whole pod.
Dry-fit	Putting parts together loosely, with no glue, just to check they fit.
Vision model / VLM	An AI that looks at a picture and describes it. The thing that actually reads the label.
Barcode fast path	If there's a barcode, look it up in a database instead of asking the AI. Faster and free.
Profile	The saved info about one user — either dosage facts, or reminders. Never both.
Guardrail	The rule stopping the AI giving medical advice. It reads labels; it never says what to take.

If you remember nothing else: the glasses hold two boxes, the boxes talk to your laptop, the laptop does the thinking, and the answer comes back as speech. Everything else on these six pages is detail underneath that one sentence.